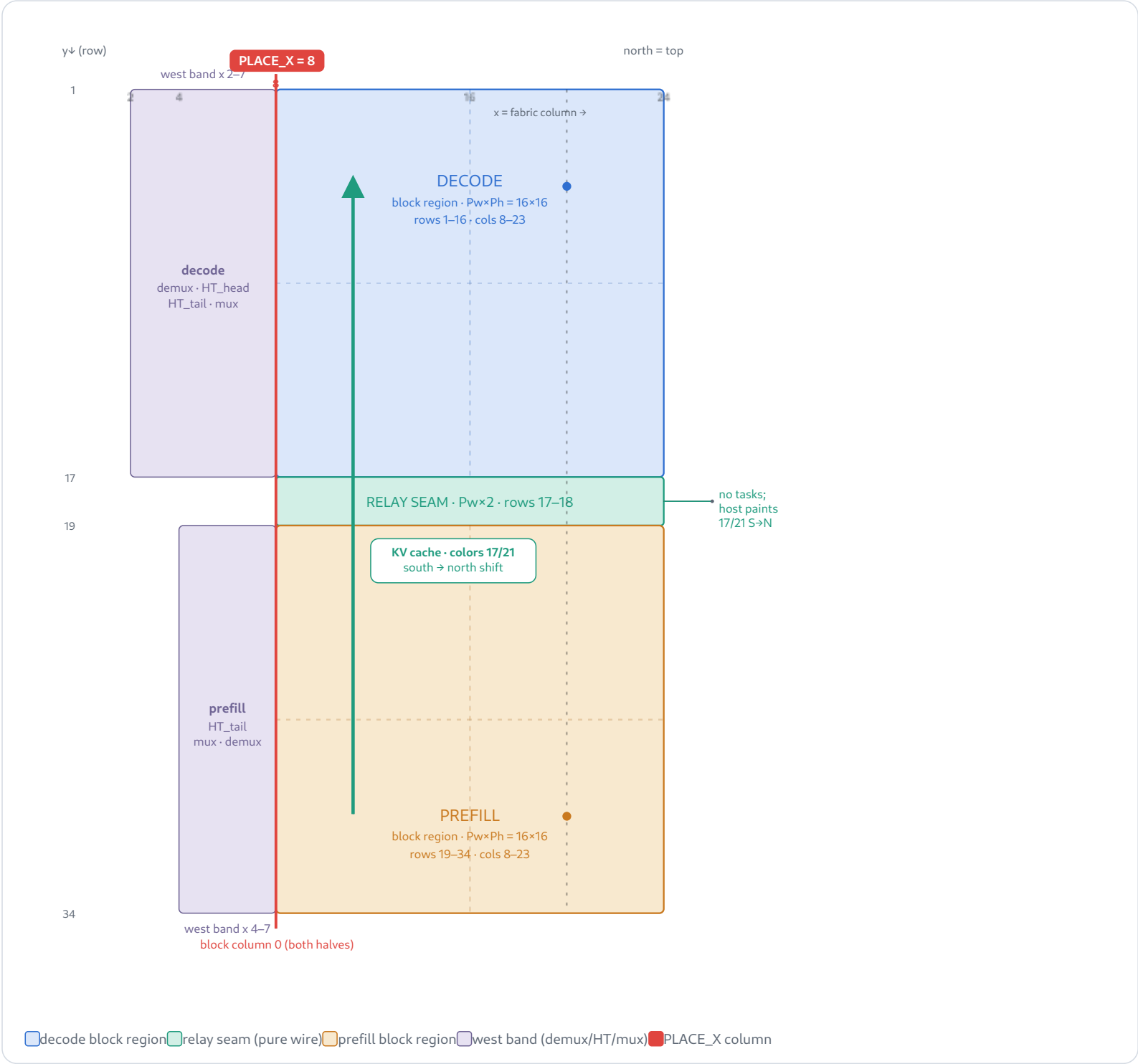


## Decode, relay seam, and prefill on one wafer

Both pipelines are compiled into a single SdkLayout artifact and stacked vertically on the fabric: decode to the north, prefill to the south, with a task-less relay seam filling the gap. The whole scheme hinges on one shared columnn, **PLACE\_X**, where both block regions begin.

drawn for **test\_sim\_2x2blk\_kv.json** · Pw=16 Ph=16 P\_BLOCK\_SIZE=8 HT\_WIDTH\_tail=4 → PLACE\_X=8



### Why PLACE\_X is the whole trick

**PLACE\_X** is the fabric X-column where **block column 0** of the block region begins. Decode and prefill are built by two separate functions that each compute their west band differently — yet both arithmetic paths are engineered to land on the *same* value, **HT\_WIDTH\_tail + 4**. Because the two block regions occupy the identical set of columns (here 8–23), decode PE at column *c* sits directly above the prefill PE at column *c* — so the KV handoff is a straight vertical (north) shift, no lateral routing.

|                                                                                                                                  |   |
|----------------------------------------------------------------------------------------------------------------------------------|---|
| <div><div>Decode</div><div>build_decode()</div><div>Has a 1-column connect strip between the HT band and the blocks.</div></div> |   |
| DEMUX_FAB_X                                                                                                                      | 2 |
| HT_HEAD_X = DEMUX+1                                                                                                              | 3 |
| HT band width (HT_WIDTH_tail)                                                                                                    | 4 |
| + 1 connect strip                                                                                                                | 1 |
| PLACE_X = 3 + 4 + 1                                                                                                              | 8 |

|                                                                                                                                                |   |
|------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <div><div>Prefill</div><div>build_prefill()</div><div>Single region — HT_tail abuts the blocks directly (manual fabric, no strip).</div></div> |   |
| HT_TAIL_X                                                                                                                                      | 4 |
| HT band width (HT_WIDTH_tail)                                                                                                                  | 4 |
| (HT_tail abuts blocks – no strip)                                                                                                              | 0 |
| PLACE_X = 4 + 4                                                                                                                                | 8 |

Two different west-band arithmetics, **one shared PLACE\_X**. The `run()` assert `HT_WIDTH_tail == P_BLOCK_SIZE/2` keeps the HT band exactly the right width so the block columns of both halves line up — the precondition for the seam to be a vertical shift.

Vertical stacking & the seam

Row placement is pure arithmetic in `run()`. Decode is pinned to the top; each region is stacked below the previous with the two-row seam in between.

| REGION                    | PLACE EXPRESSION         | FABRIC ROWS (Y) | FABRIC COLS (X) |
|---------------------------|--------------------------|-----------------|-----------------|
| <div>Decode blocks</div>  | DEC_PLACE_Y = 1          | 1 – 16          | 8 – 23          |
| <div>Decode mux</div>     | west band, seam row      | 17              | 3 – 6           |
| <div>Relay seam</div>     | RELAY_Y = 1 + Ph         | 17 – 18         | 8 – 23          |
| <div>Prefill demux</div>  | PF_PLACE_Y – 1           | 18              | 4 – 7           |
| <div>Prefill blocks</div> | PF_PLACE_Y = RELAY_Y + 2 | 19 – 34         | 8 – 23          |

- The seam is only **Pw wide** (the block columns, x 8–23) — it does *not* span the west band. At the seam rows, the west band is busy with decode's `mux` (row 17) and prefill's `demux` (row 18); the seam and those regions are X-disjoint, so they coexist on the same rows.
- `build_relay()` places a `Pw × 2` code region at `(PLACE_X, RELAY_Y)`. Its CSL body is empty — the host paints colors **17** and **21** with `input=SOUTH, output=NORTH`, turning the seam into pure prefill→decode transit wire. Color 22 stays reserved with no traffic.
- The dashed guide line in the diagram marks one representative block column *c*: the blue decode PE and the amber prefill PE on it share the same fabric X, which is exactly why the KV shift travels straight up through the seam.
- Footprint for this config: `total_w = PLACE_X + Pw + 1 = 25`, `total_h = PF_PLACE_Y + Ph + 1 = 36` — well inside the 762×1176 WSE-3 fabric.